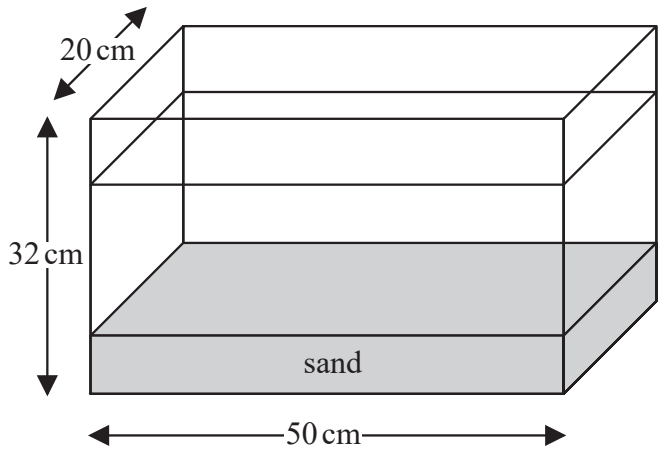
21 The diagram shows a fish tank in the shape of a cuboid.



The dimensions of the tank are 50 cm by 32 cm by 20 cm.

The tank is $\frac{3}{4}$ full of water and sand.

The ratio of the volume of water to the volume of sand is 5 : 1

Work out the number of litres of water in the tank.
You must show all your working.

Tank volume = 50 × 32 × 20 = 32000 cm^3.

1 mark
Find tank volume

Water + sand = (3/4) × 32000 = 24000 cm^3.

1 mark
Three quarters full

	Water	Sand
	5	1
×4000	20000	4000

×4000

6 parts = 24000 cm^3.

Water = 5 parts = 20000 cm^3.

1 mark
Split in ratio5:1

20000 ÷ 1000 = 20 litres.

1 mark
1000cm^3 =1litre

20 litres.

1 mark
20litres

.....litres

(Total for Question 21 is 5 marks)